

WAREHOUSE MANAGEMENT SYSTEM

**Improving warehouse operations through real-time
visibility and AI-driven insights**

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Agenda

03	CURRENT PROBLEMS
04	USE CASE DIAGRAM
06	PROPOSED SOLUTION
07	UI PROTOTYPE
08	SYSTEM WORKFLOW
09	KEY FEATURES WITH AI SOLUTIONS
15	BUSINESS BENEFITS

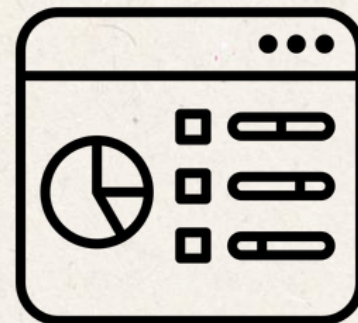
Current Problems

03

01 Stock data is scattered across multiple Excel files



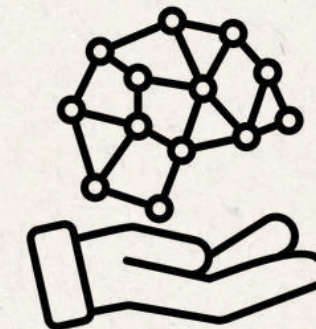
02 No real-time stock visibility and overall warehouse operations



03 Picking errors and difficulty locating items



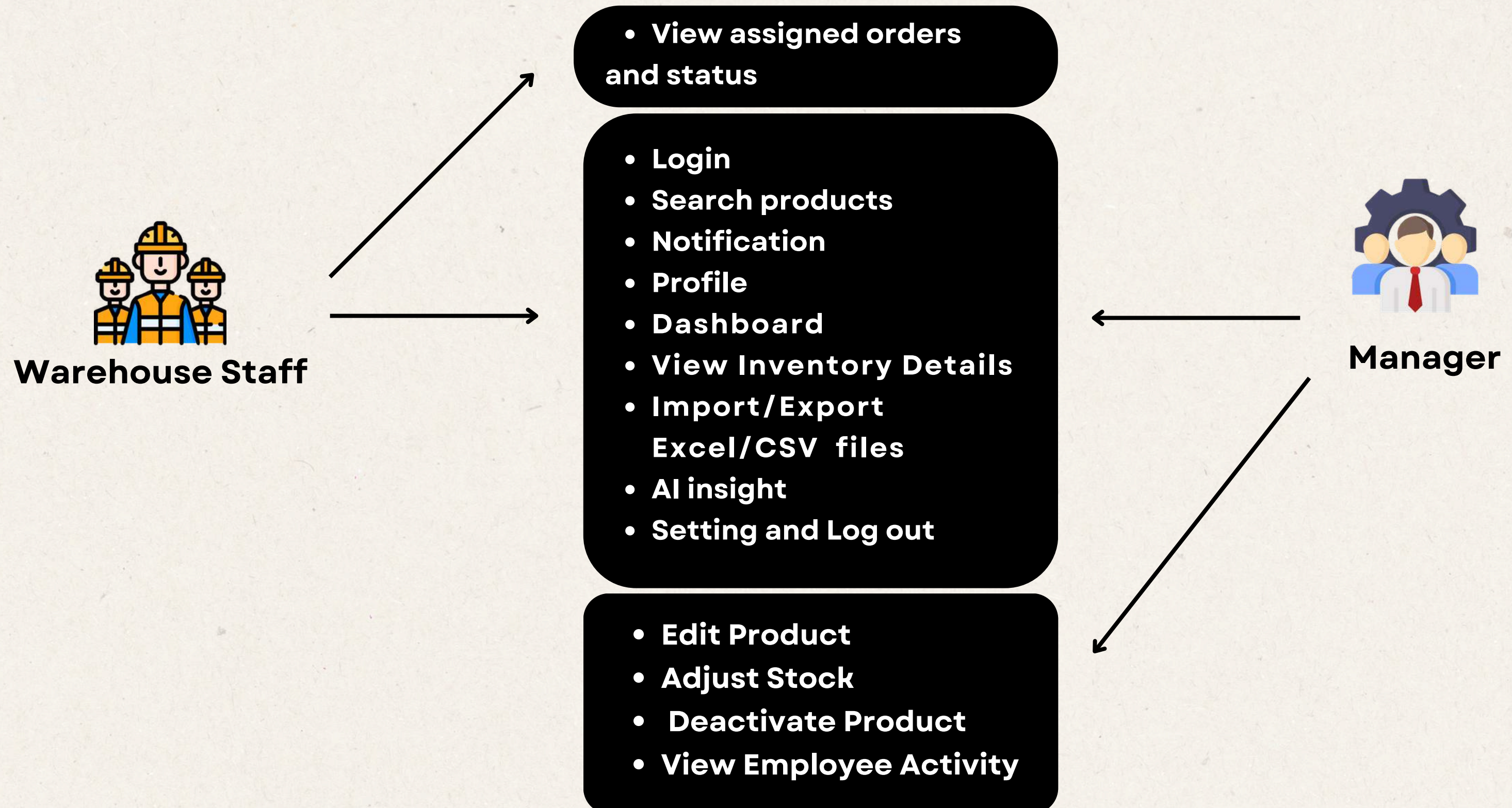
04 Limited data-driven decision support



Use Case Diagram

04

The WMS uses role-based access to provide warehouse staff with operational functions, while managers have additional monitoring and inventory management capabilities.



Features Description

Feature	Description
Login	Securely access the WMS based on user role.
Search Products	Search products using SKU, name, or location.
Notification	Receive alerts for important warehouse events such as low stock product.
Profile	View and manage user account information.
Dashboard	View an overview of inventory and warehouse operations.
View Inventory Details	Check stock quantity, location, and product status.
Import / Export Excel/CSV	Import existing inventory data and export data for reporting.
AI Insights	Provide AI-powered analysis and recommendations for warehouse operations.
View Assigned Orders & Status	Staff can view assigned picking orders and track their progress.
Edit Product, Adjust Stock, Deactivate Product	Manager can update product information and stock such as Edit name and description of product, correct stock quantities or deactivate products that are no longer in use.
View Employee Activity	Manager can monitor warehouse activities performed by employees.

Proposed Solution

06

Inventory Management

- Track inventory, stock levels and recent operations information in real time by Dashboard.
- Alert users about important warehouse events, such as low stock levels

Order & Task Tracking

- Manage orders, picking tasks, update status and provide location to avoid errors.

AI-Powered Insights

- Use AI to forecast demand, optimize picking and product placement, predict stockouts, and detect inventory anomalies.

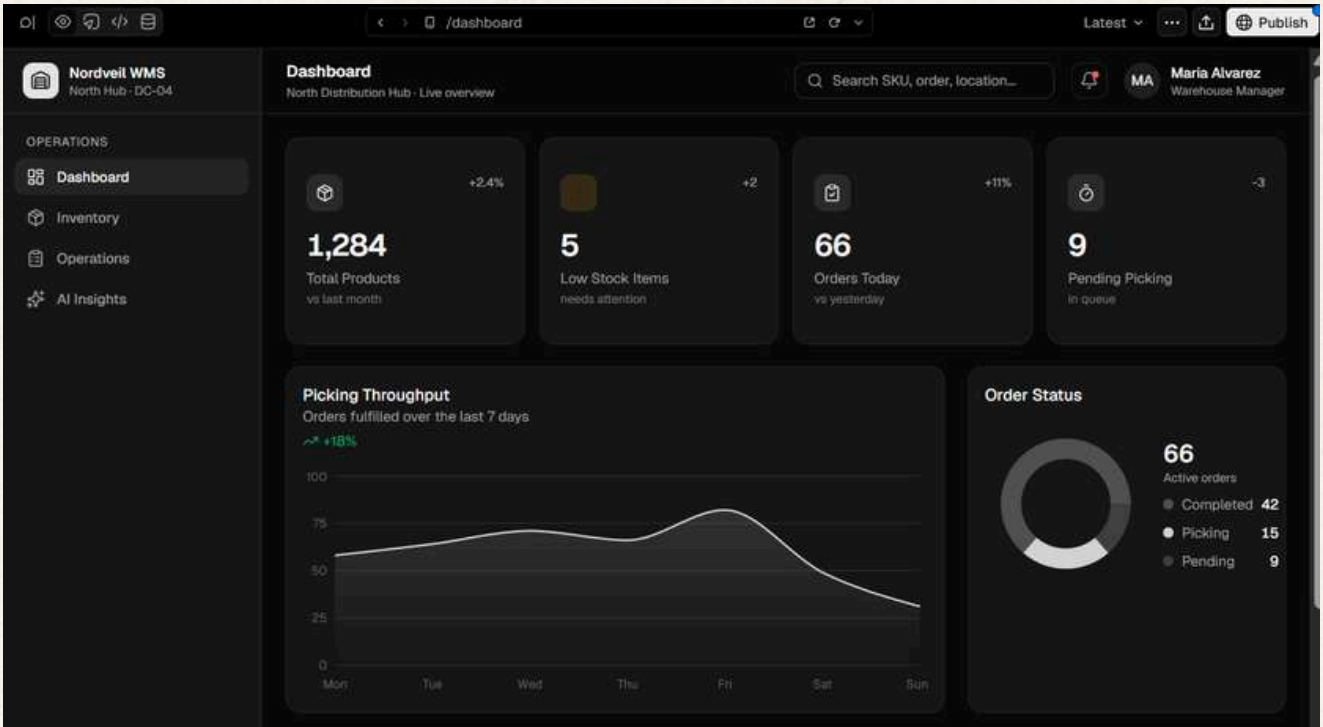
Role-Based Access Control

- Provide secure access based on user roles with appropriate permissions.

Excel Data Migration & Import

- Transfer existing inventory data from Excel files into the new WMS, enabling a smooth transition from the current process without losing historical data.

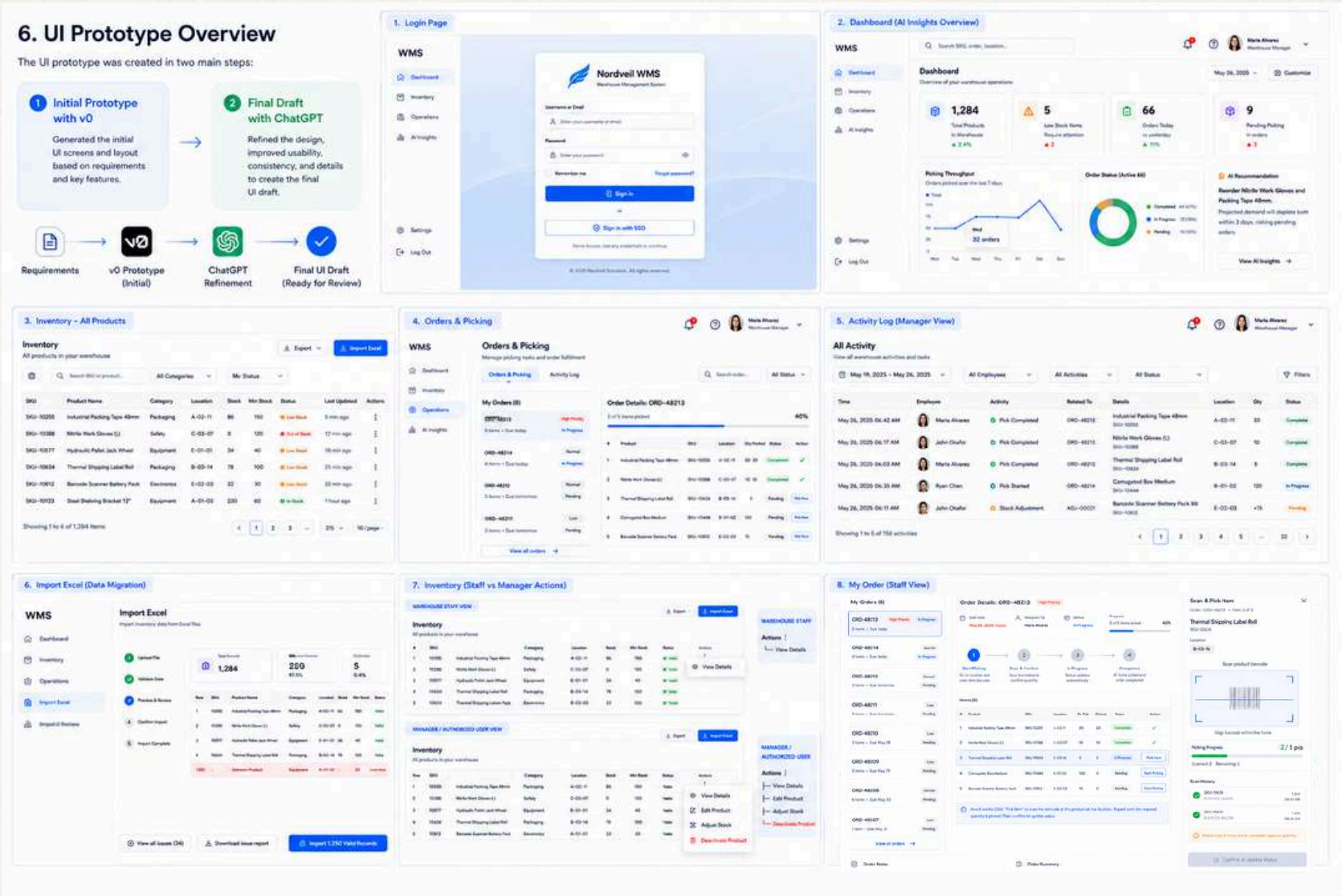
UI PROTOTYPE



1st Drafts

Used v0 to quickly generate the initial WMS interface based on the defined system requirements and key features.

V0 Link

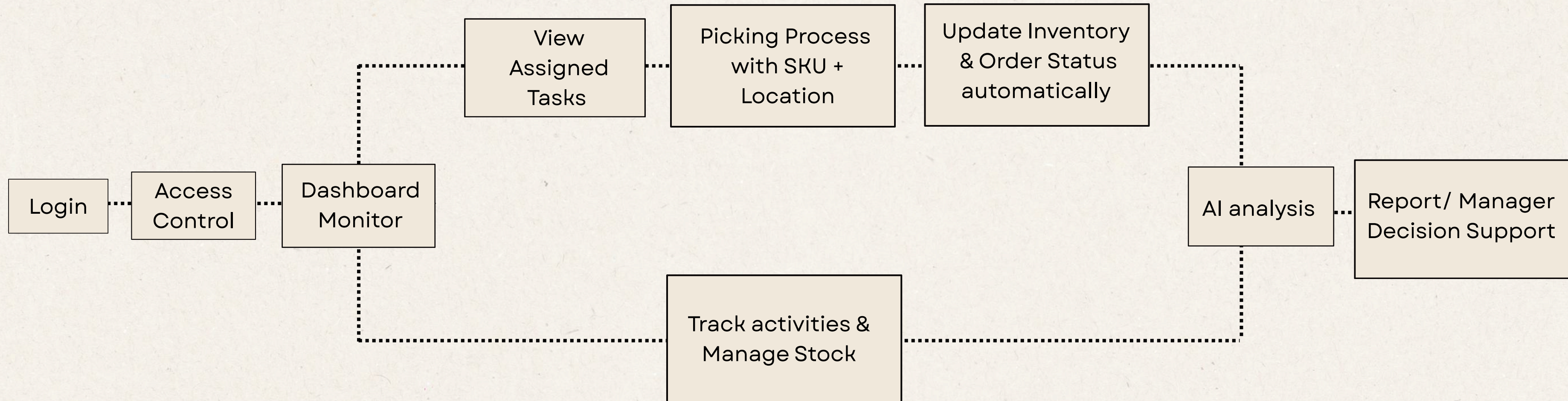


Final Drafts

Used ChatGPT to refine the UI, adjust individual screens, and create the final design draft with consistent layouts and functionality.

System Workflow

08

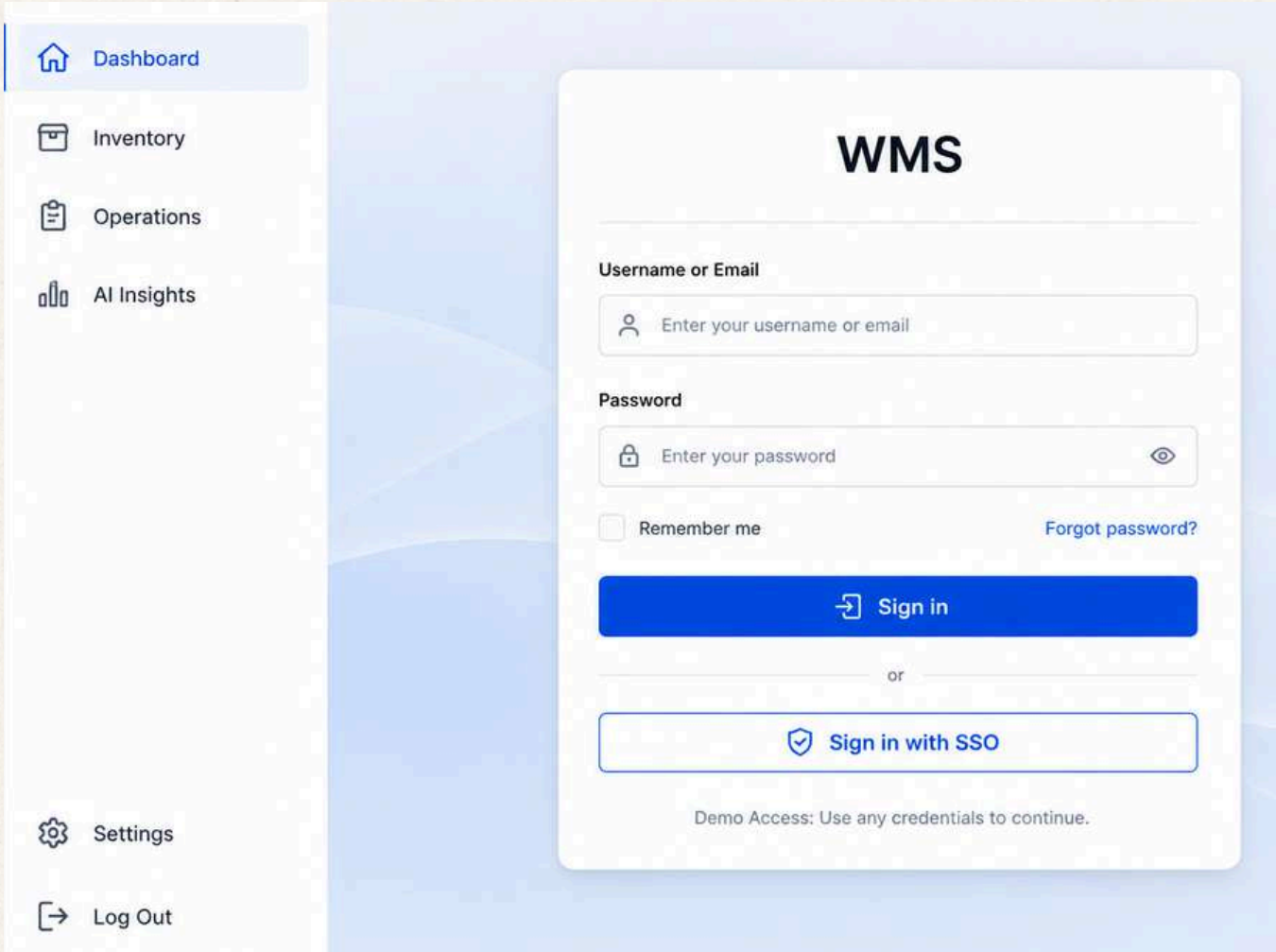


KEY FEATURES WITH AI SOLUTIONS

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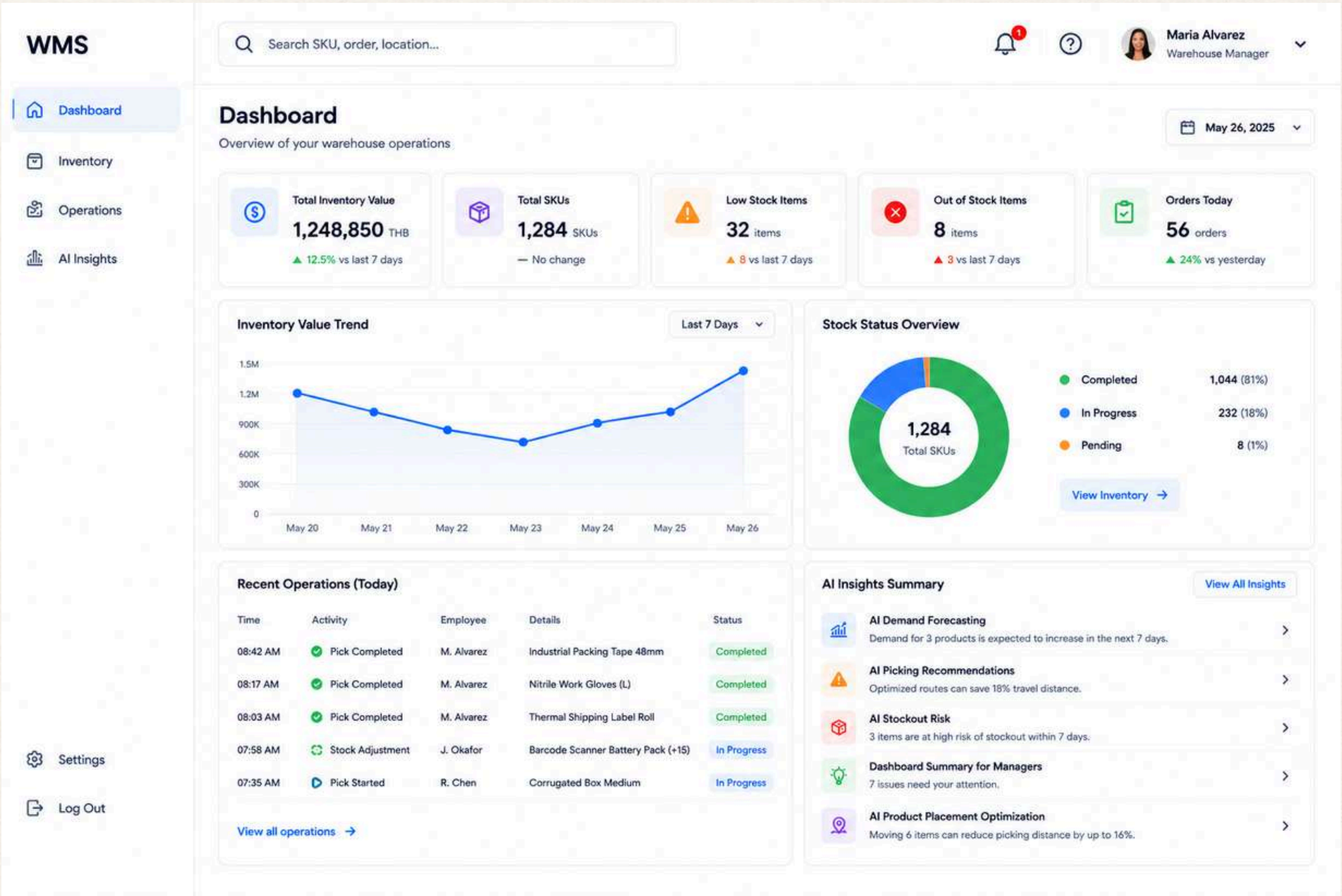
1. Login Page

- Allows users to securely access the WMS system using their credentials.



2.Home page/Dashboard

- Provides an overview of warehouse operations, including inventory status, orders, recent activities, and key performance indicators.
- Alerts users when stock falls below the minimum level or when orders are overdue and incomplete.



KEY FEATURES WITH AI SOLUTIONS

3.Inventory

- Displays real-time inventory information and allows authorized users to view, update inventory data

Staff view :

Inventory

All products in your warehouse

Export

Import Excel

Search SKU or product...

All Categories

All Status

More Filters

SKU	Product Name	Category	Location	Stock	Min Stock	Status	Last Updated	Actions
SKU-10255	Industrial Packing Tape 48mm	Packaging	A-02-11	86	150	Low Stock	5 min ago	⋮
SKU-10388	Nitrile Work Gloves (L)	Safety	C-03-07	0	120	Out of Stock	12 min ago	⋮
SKU-10577	Hydraulic Pallet Jack Wheel	Equipment	E-01-01	34	40	Low Stock	18 min ago	⋮
SKU-10634	Thermal Shipping Label Roll	Packaging	B-03-14	78	100	Low Stock	25 min ago	⋮
SKU-10812	Barcode Scanner Battery Pack	Electronics	E-02-03	22	30	Low Stock	32 min ago	⋮
SKU-10123	Steel Shelving Bracket 12"	Equipment	A-01-03	230	60	In Stock	1 hour ago	⋮

Showing 1 to 6 of 1,284 items

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1

2

3

...

215

10 / page

View Details

Manager view :

Inventory

All products in your warehouse

Export

Import Excel

Search SKU or product...

All Categories

All Status

More Filters

SKU	Product Name	Category	Location	Stock	Min Stock	Status	Last Updated	Actions
SKU-10255	Industrial Packing Tape 48mm	Packaging	A-02-11	86	150	Low Stock	5 min ago	⋮
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SKU-10123	Steel Shelving Bracket 12"	Equipment	A-01-03	230	60	In Stock	1 hour ago	⋮

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10 / page

View Details

Edit Product

Adjust Stock

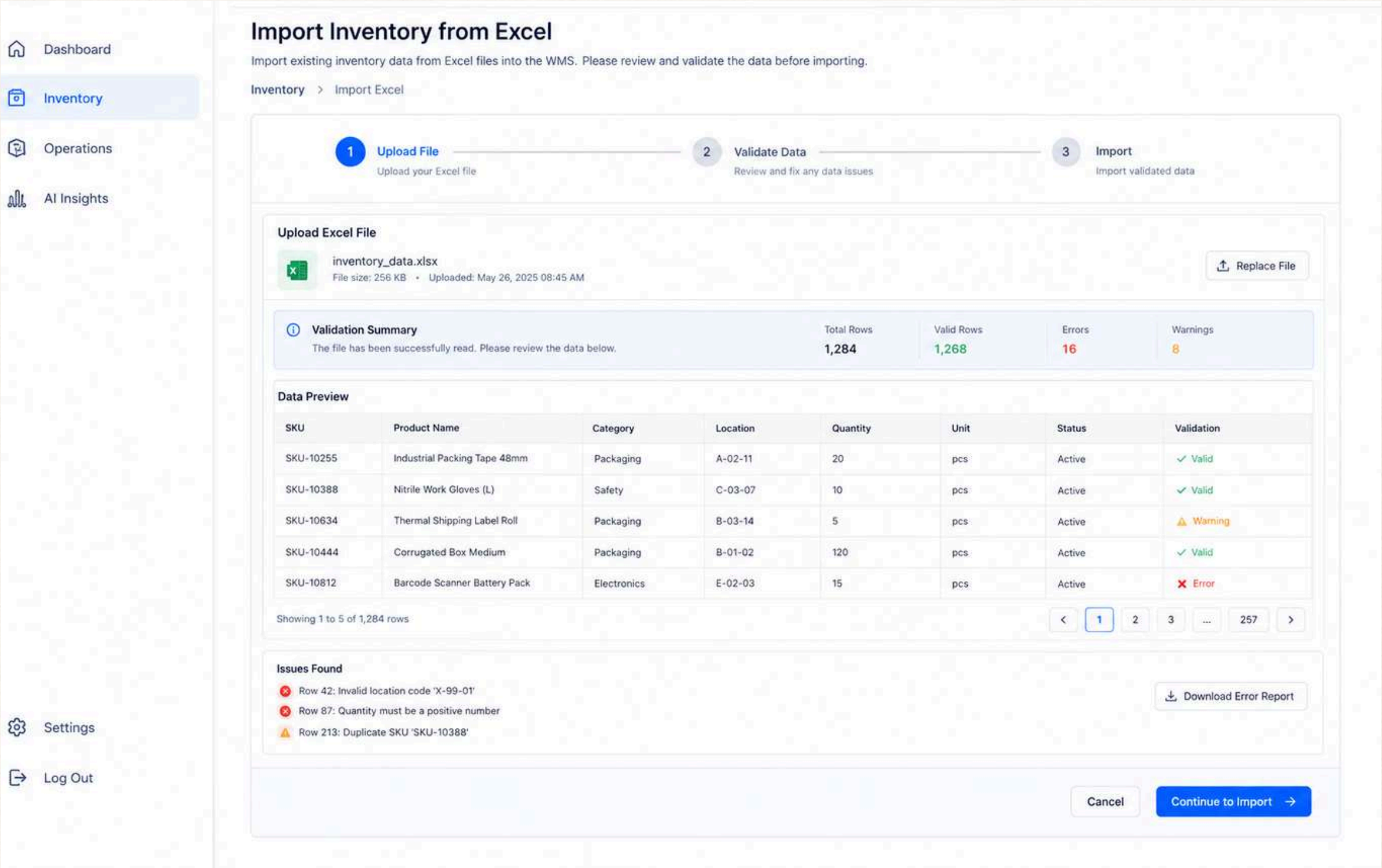
Deactivate Product

Additional features for authorized user

KEY FEATURES WITH AI SOLUTIONS

3.Inventory

- Allows existing inventory data from Excel files to be transferred into the WMS, with data validation before importing
- Supports Excel/csv files export for report or analysis



KEY FEATURES WITH AI SOLUTIONS

4.Operations

- Shows assigned picking orders, item locations, quantities, priorities, and picking progress to help warehouse staff complete tasks efficiently.
- Stock and order status are updated automatically in real time via scanning Barcode / SKU during warehouse operations to reduce human errors.

Staff view:

My Orders (8)

ORD-48213

High Priority

In Progress

5 items • Due today

ORD-48214

Normal

In Progress

4 items • Due today

ORD-48212

Normal

Pending

3 items • Due tomorrow

ORD-48211

Low

Pending

2 items • Due tomorrow

ORD-48210

Low

Pending

3 items • Due May 28

ORD-48209

Low

Pending

2 items • Due May 29

ORD-48208

Normal

Pending

6 items • Due May 30

ORD-48207

Low

Pending

1 item • Due May 31

View all orders →

Order Details: ORD-48213

High Priority

Due Date

May 26, 2025 (Today)

Assigned To

Maria Alvarez

Status

In Progress

Progress

2 of 5 items picked

40%

1

2

3

4

Start Picking

Go to location and scan item barcode

Scan & Confirm

Scan barcode and confirm quantity

In Progress

Status updates automatically

Completed

All items picked and order completed

Items (5)

#	Product	SKU	Location	To Pick	Picked	Status	Action
1	Industrial Packing Tape 48mm	SKU-10255	A-02-11	20	20	Completed	✓
2	Nitrile Work Gloves (L)	SKU-10388	C-03-07	10	10	Completed	✓
3	Thermal Shipping Label Roll	SKU-10634	B-03-14	5	2	In Progress	Pick Item
4	Corrugated Box Medium	SKU-10444	B-01-02	120	0	Pending	Start Picking
5	Barcode Scanner Battery Pack	SKU-10812	E-02-03	15	0	Pending	Start Picking

How it works:

Click "Pick Item" to scan the barcode of the product at the location. Repeat until the required quantity is picked. Then confirm to update status.

Order Notes

Handle with care. Ensure all items are packed securely.

Order Summary

Total Items	Total Quantity	Picked	Remaining
5	170	30 (40%)	140

Scan & Pick Item

Order: ORD-48213 • Item 3 of 5

Thermal Shipping Label Roll

SKU-10634

Location

B-03-14

Scan product barcode

Align barcode within the frame

Picking Progress

2 / 5 pcs

Scanned: 2 Remaining: 3

Scan History

✓

SKU-10634

8 859123 456789

1 pcs

08:45 AM

✓

SKU-10634

8 859123 456789

1 pcs

08:46 AM

⚠

Please scan 3 more pcs to complete required quantity.

Confirm & Update Status

Cancel

Warehouse staff scan the product SKU barcode during picking. The system automatically updates the picked quantity after each scan, and the order can only be confirmed once the required quantity is fully scanned.

KEY FEATURES WITH AI SOLUTIONS

4.Operations

- Provides managers with visibility into warehouse activities, including employee actions, order progress & details, stock adjustments, and activity status.

Dashboard

Inventory

Operations

AI Insights

Settings

Log Out

All Activity

View all warehouse activities and tasks

May 19, 2025 - May 26, 2025

All Employees

All Activities

All Status

Filters

Time	Employee	Activity	Related To	Details	Location	Qty	Status
May 26, 2025 06:42 AM	 Maria Alvarez	 Pick Completed	ORD-48213	Industrial Packing Tape 48mm SKU-10255	A-02-11	20	Complete
May 26, 2025 06:17 AM	 John Okafor	 Pick Completed	ORD-48213	Nitrile Work Gloves (L) SKU-10388	C-03-07	10	Complete
May 26, 2025 06:03 AM	 Maria Alvarez	 Pick Completed	ORD-48213	Thermal Shipping Label Roll SKU-10634	B-03-14	5	Complete
May 26, 2025 06:35 AM	 Ryan Chen	 Pick Started	ORD-48214	Corrugated Box Medium SKU-10444	B-01-02	120	In Progress
May 26, 2025 06:11 AM	 John Okafor	 Stock Adjustment	ADJ-00021	Barcode Scanner Battery Pack SKU-10812	E-02-03	+15	Pending

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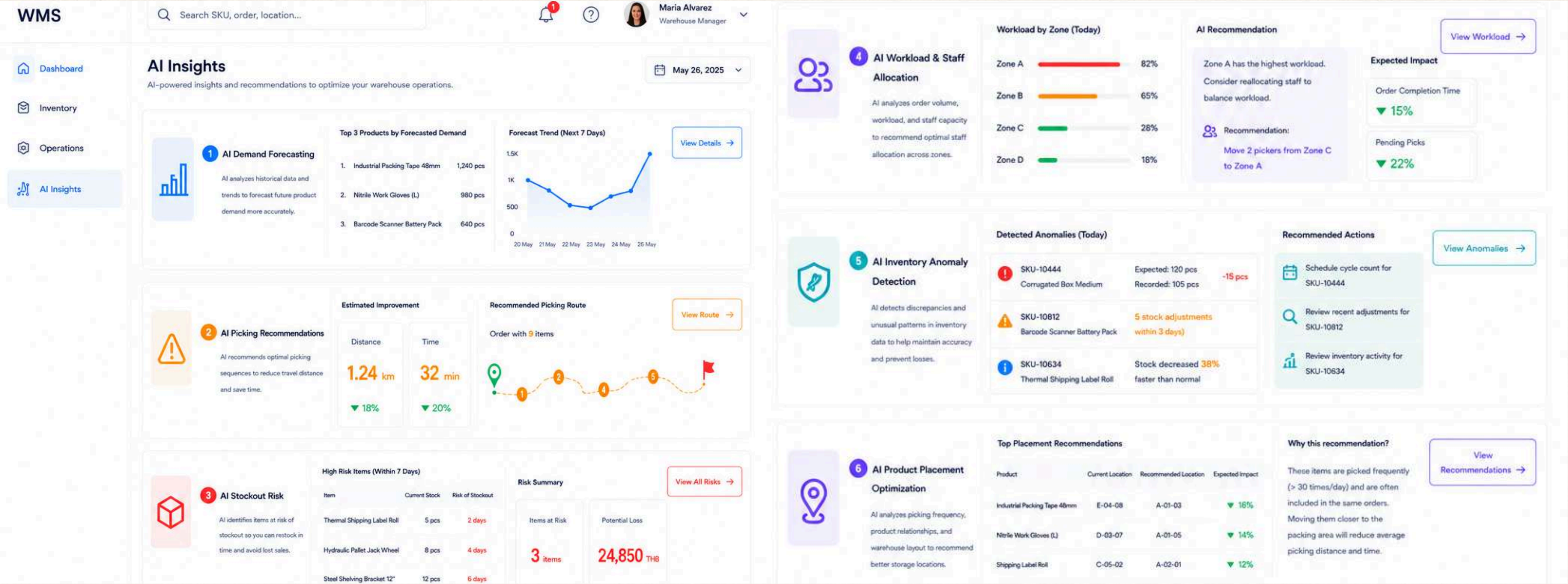
Manager view

KEY FEATURES WITH AI SOLUTIONS

14

5.AI insight

- Uses warehouse data to provide AI-powered insights such as demand forecasting, picking recommendations, stockout risks, workload allocation, anomaly detection, and product placement optimization.



Business Benefits

15

01 Improve Operational Efficiency

- Reduce data lost and time checking multiple Excel files by centralizing warehouse data in one system.

02 Reduce Human Errors

- Barcode/SKU scanning automatically updates inventory and status, reducing errors caused by incorrect stock
- updates and also help warehouse staff locate products and complete picking tasks faster with clear order and location information.

03 Improve Inventory Visibility and Management Decisions

- Provide real-time visibility of stock levels, product locations, order status, and warehouse activities to support faster and more data-driven decisions.

04 Optimize Warehouse Resources by AI

- Use inventory monitoring and AI demand forecasting to identify stock risks and support better replenishment planning.
- AI recommendations can help improve picking routes, staff allocation, and product placement to reduce unnecessary time and movement.

05 Easy Migration & Reporting

- Existing inventory data can be transferred into the WMS, while data can also be exported to Excel/CSV for reporting and further analysis.